

PER2-PROTAC: Data exploration and statistical testing

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2026-08-11

Here, we explore the protein-level quantitative data and perform statistical testing to identify phosphorylation changes upon PROTAC treatment.

Setup

Load required packages

Define directory for plots

```
plots_dir <- here('results/plots')
if(!dir.exists(plots_dir)){
  dir.create(plots_dir, recursive = TRUE)
}
```

Read in the data

```
output_dir <- here('notebooks/output')

qfs <- readRDS(here(output_dir, 'PROTAC_qfs.rds'))

long_format_protein <- readRDS(
  here(output_dir, 'PROTAC_long_format_protein.rds'))

long_long_format_phosphopeptide_norm_to_protein <- readRDS(
  here(output_dir, 'PROTAC_long_long_format_phosphopeptide_norm_to_protein.rds'))
```

Data exploration

PCA plots

Below, we make PCA plots for the protein and phosphopeptide data. At the protein level, DMSO and PROTAC separate on PC1 (30% of variance), with two DMSO-2 and PROTAC-3 relatively close together. For phosphopeptides, the conditions separate very well (PC1; ~40% variance explained)

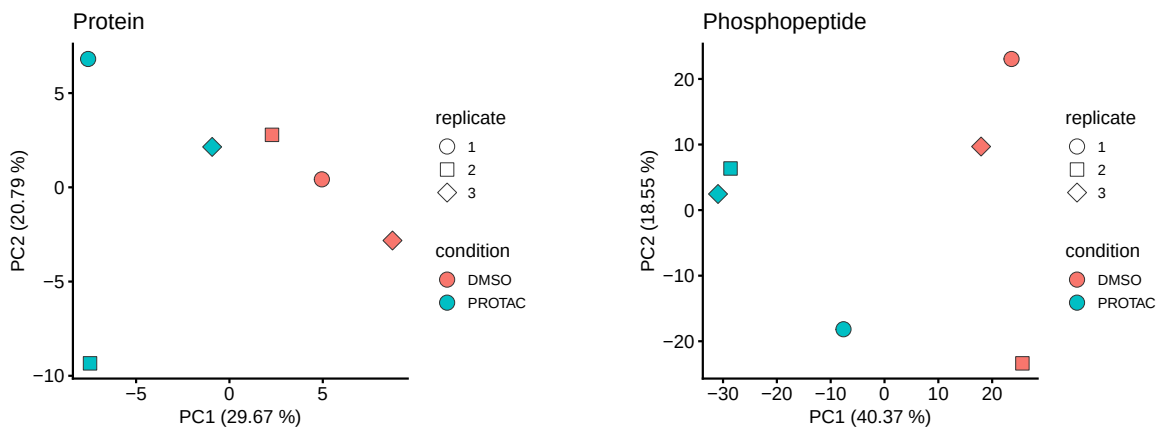
```
for(qf in names(qfs)){
  if(qf=='total'){
    assay_name <- 'protein'
  }
}
```

```

    feature <- 'Protein'
  } else{
    assay_name <- 'phosphopeptide'
    feature <- 'Phosphopeptide'
  }

  print(plot_pca(
    qfs[[qf]], i=assay_name, colour_by = 'condition', shape='replicate') +
    ggtitle(feature))
}

```



Statistical testing of protein abundance changes between conditions

Protein

First, we'll perform statistical testing to compare the conditions (PROTAC vs DMSO) for total protein abundance

```

## Create design matrix for the linear model
condition <- factor(qfs$total[['protein']]$condition, levels=c('DMSO', 'PROTAC'))
m_design <- model.matrix(~condition)

## Fit linear model using the design matrix and desired contrasts
fit_model <- lmFit(object = assay(qfs$total[['protein']]), design = m_design)

## Update the model using the limma eBayes algorithm
final_model <- eBayes(fit = fit_model,
                      trend = TRUE,
                      robust = TRUE)

## Add TREAT to final model to perform fold-change thresholding for contrasts
# require |FC| significantly over 1.25x
final_model_treat <- treat(fit_model, fc=1.25, trend=TRUE, robust=TRUE)

## QC plot to assess if there is a trend between abundance and variance, as expected

```

```

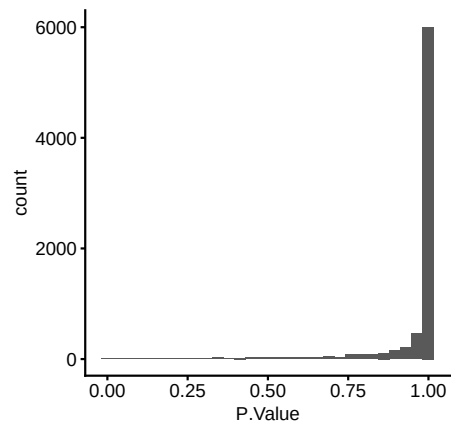
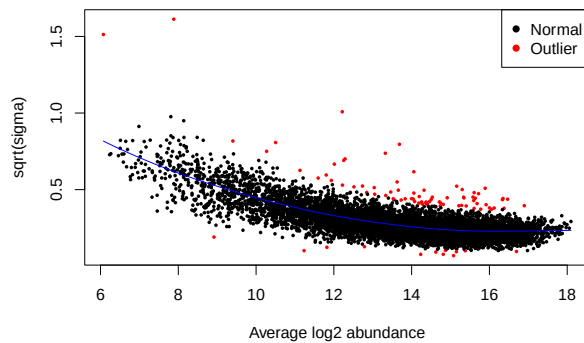
plotSA(fit = final_model,
       cex = 0.5,
       xlab = "Average log2 abundance")

## Format results
prot_limma_results <- topTreat(
  fit = final_model_treat,
  coef = 'conditionPROTAC',
  adjust.method = "BH", # Method for multiple hypothesis testing
  number = Inf) %>% # Print results for all proteins
  tibble::rownames_to_column("Protein") %>%
  merge(data.frame(rowData(qfs$total)[['protein']] ),
        by.x='Protein', by.y='row.names', all.x=TRUE) %>%
  arrange(P.Value)

## Check distribution of p-values. Because we are using treat, we don't expect a
## uniform distribution under the null, but rather a peak near 1.
p <- prot_limma_results %>%
  as_tibble() %>%
  ggplot(aes(x = P.Value)) +
  geom_histogram() +
  theme_biomasslmb()

print(p)

```



How many proteins have a significant change in abundance

```

table(prot_limma_results$adj.P.Val<0.05,
      ifelse(prot_limma_results$logFC>0, 'Increased', 'Decreased')) %>%
  knitr::kable()

```

	Decreased	Increased
FALSE	3586	4060
TRUE	1	1

```
prot_limma_results %>%
  filter(adj.P.Val<0.05) %>%
  select(Protein, Gene.Names.First, logFC, adj.P.Val) %>%
  knitr::kable()
```

Protein	Gene.Names.First	logFC	adj.P.Val
O00469	PLOD2	-0.8884707	7.00e-07
Q16850	CYP51A1	0.7847998	4.25e-05

OK, so very few proteins have a significant change in abundance!

Phosphopeptide statistical testing

Next, we perform statistical testing for the phosphopeptide data, comparing PROTAC vs DMSO while accounting for the matched protein abundance measurements.

```
# Create data matrix for modelling
to_model_phospho <- long_long_format_phosphopeptide_norm_to_protein %>%
  filter(type!='phosphopeptide_prot_norm') %>%
  select(prot_phos, condition, replicate, type, abundance) %>%
  pivot_wider(names_from = c(condition, replicate, type), values_from=abundance) %>%
  tibble::column_to_rownames('prot_phos')

# Create design matrix for the linear model
exp_cond_phospho <- data.frame(sample=colnames(to_model_phospho)) %>%
  separate(sample, into=c('condition', 'replicate', 'type'), remove = FALSE)

condition <- factor(exp_cond_phospho$condition, levels=c('DMSO', 'PROTAC'))

type <- factor(exp_cond_phospho$type, levels=c('protein', 'phosphopeptide'))

m_design_phospho <- model.matrix(~ condition * type)

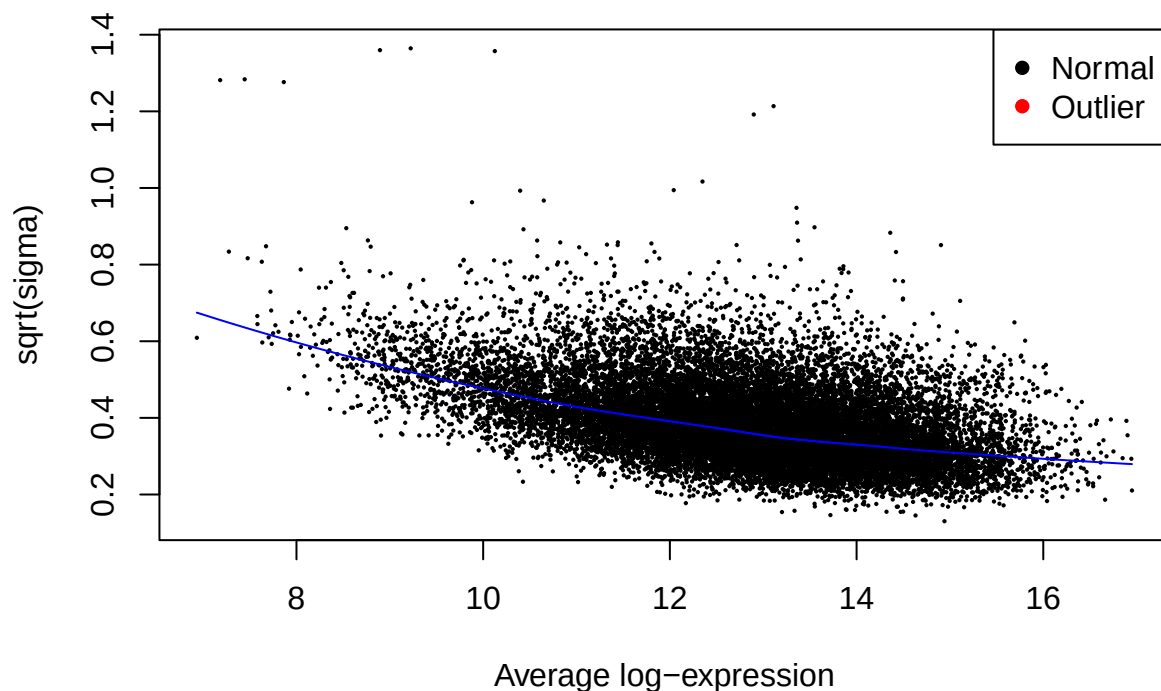
# same ID for phospho + total from the same sample
block <- interaction(condition, exp_cond_phospho$replicate)

# Estimate the correlation between measurements from the same block
corfit <- duplicateCorrelation(as.matrix(to_model_phospho),
                              m_design_phospho, block=block)

# Fit the linear model with blocking and correlation information
limma_fit_phospho <- lmFit(
  as.matrix(to_model_phospho), m_design_phospho, block=block, correlation=corfit$consensus)

# Update the model using the limma eBayes algorithm
limma_fit_phospho_eb <- eBayes(limma_fit_phospho, trend=TRUE, robust=TRUE)

# QC plot to assess if there is a trend between abundance and variance, as expected,
# and if the limma 'trend' captures this
plotSA(limma_fit_phospho_eb)
```



```
# Format results
phospho_limma_results <- topTable(limma_fit_phospho_eb, coef = 'conditionPROTAC:typephosphopeptide',
  adjust.method = "BH",      # Method for multiple hypothesis testing
  number = Inf) %>%         # Print results for all proteins
  tibble::rownames_to_column("phosphosite") %>%
  merge(rowData(qfs$phospho)[['phosphopeptide']][
    ,c('Entry.Name', 'Protein.names.y', 'Gene.Names', 'Gene.Names.First', 'site_seq')],
    by.x='phosphosite', by.y='row.names', all.x=TRUE) %>%
  data.frame() %>%
  separate(phosphosite, into=c('UniprotID', 'phospho_pos'), sep='_', remove=FALSE)
```

Save results

```
saveRDS(phospho_limma_results, here(output_dir, 'PROTAC_phospho_limma_results.rds'))
```

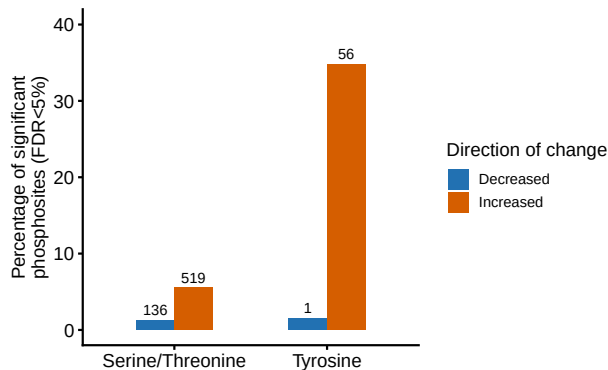
How many phosphosites have a significant change in abundance?

Below, we summarise how many phosphosites have a significant change in abundance (FDR<5%)

```
table(ifelse(phospho_limma_results$adj.P.Val<0.05, 'Significant', 'Not significant'),
  ifelse(phospho_limma_results$logFC>0, 'Increased', 'Decreased')) %>%
  knitr::kable()
```

	Decreased	Increased
Not significant	10377	9007
Significant	137	575

```
phospho_limma_results %>%
  mutate(type=ifelse(grepl('Y', phospho_pos), 'Tyrosine', 'Serine/Threonine')) %>%
  mutate(sig=adj.P.Val<0.05) %>%
  group_by(sig, type, direction=ifelse(logFC>0, 'Increased', 'Decreased')) %>%
  tally() %>%
  group_by(type, direction) %>%
  mutate(perc=100*n/sum(n)) %>%
  filter(sig) %>%
  ungroup() %>%
  complete(direction, type, fill=list(n=0)) %>%
  ggplot(aes(type, perc, fill=direction)) +
  geom_col(position='dodge', width=0.5) +
  theme_biomasslmb() +
  labs(x='', y='Percentage of significant\nphosphosites (FDR<5%)', fill='Direction of change') +
  scale_fill_manual(values=get_cat_palette(2)) +
  geom_text(aes(label=n), position=position_dodge(width=0.5), vjust=-0.5, size=4) +
  ylim(0, 40) +
  theme(strip.background = element_blank())
```



Below, we make a volcano plot for the statistical test results

```
# highlight top 10 significant proteins per genotype
phospho_limma_results %>%
  group_by(logFC>0) %>%
  filter(adj.P.Val<0.05) %>%
  slice_min(P.Value, n=10) -> highlight

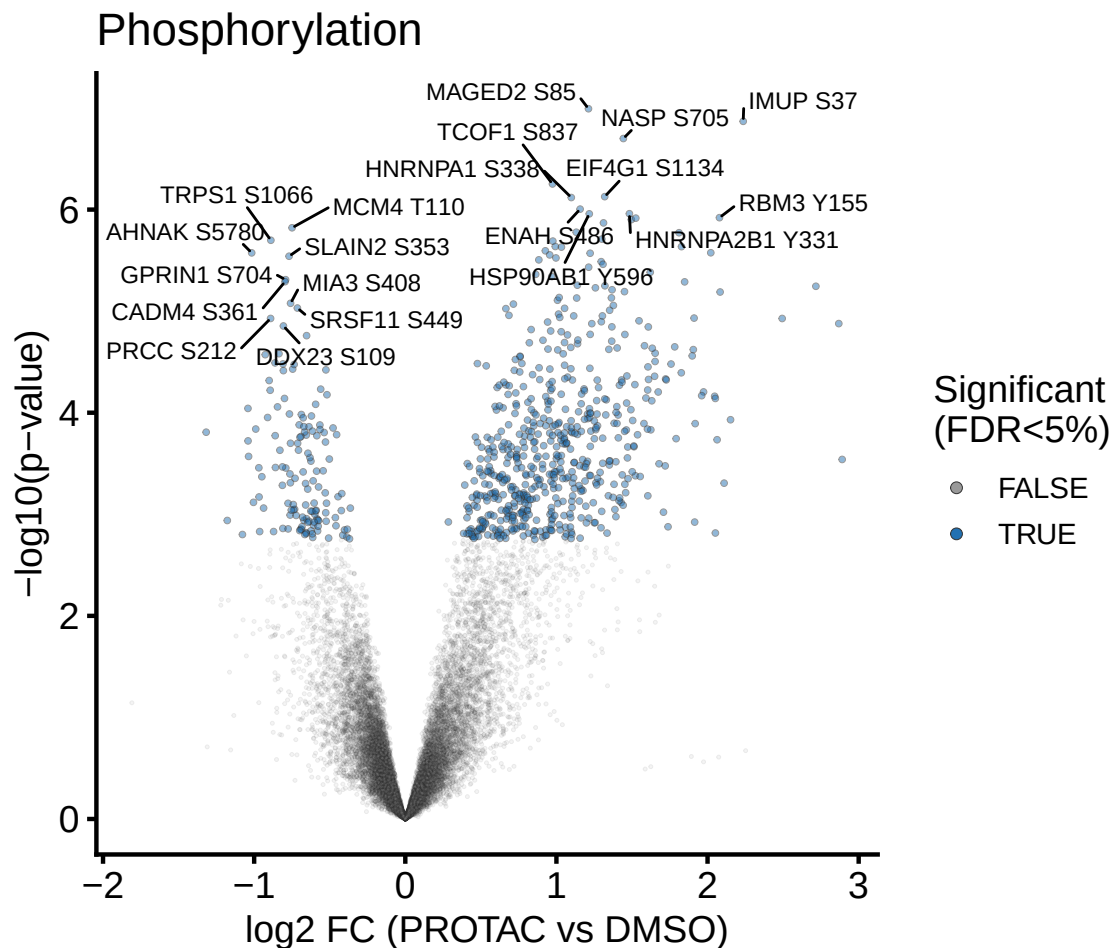
phospho_limma_results %>%
  mutate(significant=adj.P.Val<0.05) %>%
  ggplot(aes(logFC, -log10(P.Value))) +
  geom_point(aes(fill=significant,
                 alpha=significant, size=significant),
            colour='grey20', stroke=0.25, shape=21) +
  theme_biomasslmb() +
```

```

scale_fill_manual(values=c('grey60', get_cat_palette(1)),
                  name='Significant\n(FDR<5%)' ) +
scale_size_manual(values=c(0.5, 1), guide='none') +
scale_alpha_manual(values=c(0.1, 0.5), guide='none') +
labs(x='log2 FC (PROTAC vs DMSO)', y='-\log10(p-value)') +
theme(strip.background = element_blank()) +
geom_text_repel(
  data=highlight,
  aes(label=paste(Gene.Names.First, gsub('.*_', '', phosphosite))),
  max.overlaps = 50, size=3, min.segment.length =0, colour='black', force = 10) +
ggtitle('Phosphorylation') -> p

print(p)

```



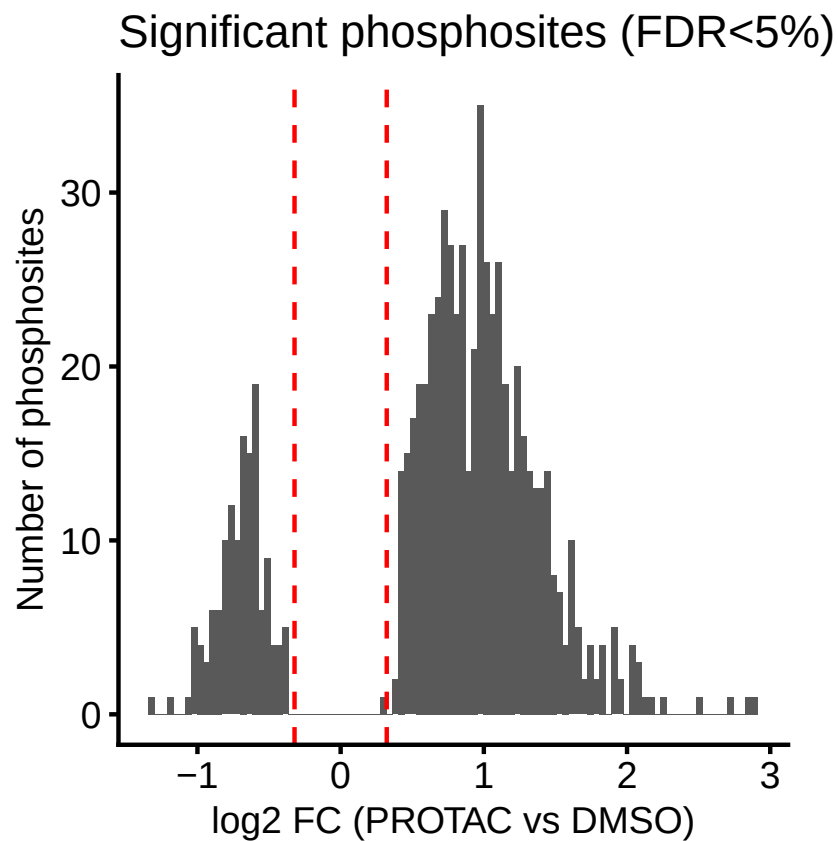
```

table(dir=ifelse(phospho_limma_results$logFC>0, 'inc', 'dec'), sig=phospho_limma_results$adj.P.Val<0.05

```

```
## , , over_1.1 = FALSE
##
##      sig
## dir   FALSE TRUE
##  dec  5427    0
##   inc  4321    0
##
## , , over_1.1 = TRUE
##
##      sig
## dir   FALSE TRUE
##  dec  4950   137
##   inc  4686   575
```

```
phospho_limma_results %>%
  filter(adj.P.Val<0.05) %>%
  ggplot(aes(logFC)) +
  geom_histogram(bins=100) +
  theme_biomasslmb() +
  labs(x='log2 FC (PROTAC vs DMSO)', y='Number of phosphosites') +
  theme(strip.background = element_blank()) +
  ggtitle('Significant phosphosites (FDR<5%)') +
  geom_vline(xintercept = log2(1.25), colour='red', linetype='dashed') +
  geom_vline(xintercept = -log2(1.25), colour='red', linetype='dashed')
```



Plot changes in phosphorylation for AKT1S1

Below, we plot the changes in phosphorylation for AKT1S1, which has multiple significantly changing sites

```
p_values <- phospho_limma_results %>%
  filter(Gene.Names.First=='AKT1S1') %>%
  mutate(ptm_name=phosphosite)

to_plot <- long_long_format_phosphopeptide_norm_to_protein %>%
  filter(Gene.Names.First=='AKT1S1', type=='phosphopeptide_prot_norm')

DMSO_abundances <- to_plot %>%
  filter(condition=='DMSO') %>%
  group_by(prot_phos) %>%
  summarise(abundance=mean(abundance))

to_plot <- to_plot %>%
  merge(DMSO_abundances, by='prot_phos', suffix=c('_', '_DMSO')) %>%
  mutate(abundance_diff=abundance - abundance_DMSO) %>%
  group_by(condition, prot_phos) %>%
  mutate(mean_abundance_diff=mean(abundance_diff)) %>%
  arrange(desc(mean_abundance_diff)) %>%
  mutate(ptm_name_facet=gsub(';', ' ', '\n', ptm_name))

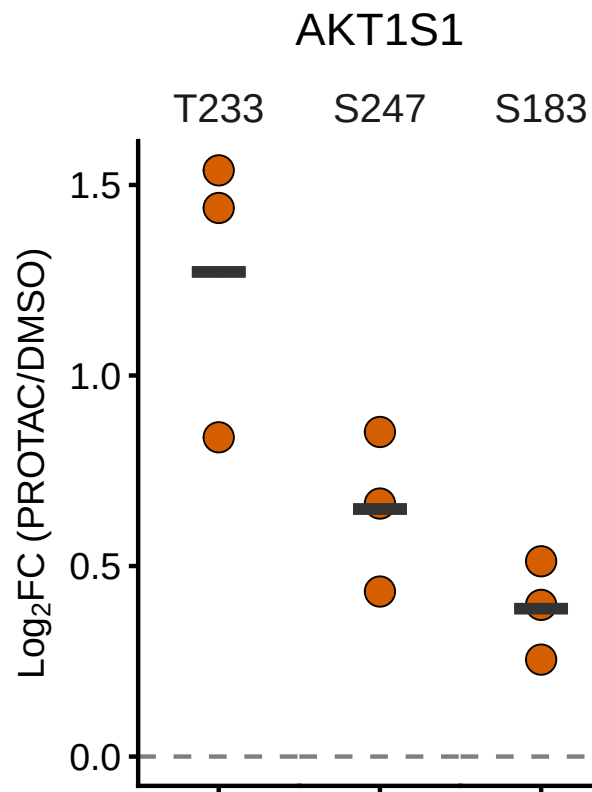
to_plot$ptm_name_facet <- factor(to_plot$ptm_name_facet,
                                levels=unique(to_plot$ptm_name_facet))

p_values <- phospho_limma_results %>%
  filter(Gene.Names.First=='AKT1S1')

sig_sites <- p_values %>%
  filter(adj.P.Val<0.1) %>%
  pull(phospho_pos)

p <- to_plot %>%
  filter(ptm_name %in% sig_sites) %>%
  filter(condition!='DMSO') %>%
  ggplot(aes(ptm_name, abundance_diff)) +
  geom_hline(yintercept = 0, colour='grey50', linetype='dashed') +
  geom_point(colour='black', stroke=0.5, shape=21,
             fill=get_cat_palette(2)[2], size=5) +
  stat_summary(fun.min=mean, fun.max=mean, geom='errorbar',
              linewidth=2, width=0.4, colour='grey20') +
  theme_biomasslmb(aspect_square = FALSE) +
  labs(x='', y=expression(Log[2]*FC~"(PROTAC/DMSO)")) +
  theme(axis.text.x=element_text(angle=45, hjust=1, vjust=1)) +
  facet_grid(.~ptm_name_facet, scales='free_x') +
  theme(strip.background = element_blank(),
        panel.spacing = unit(0, 'mm'),
        axis.text.x = element_blank(),
        aspect.ratio=4,
        strip.text.x=element_text(vjust=1, hjust=0),
        plot.title = element_text(hjust=0.5)) +
  ggtitle('AKT1S1')
```

```
p1 <- p + theme(strip.text.x=element_text(vjust=0.5, hjust=0.5, size=15))
print(p1)
```



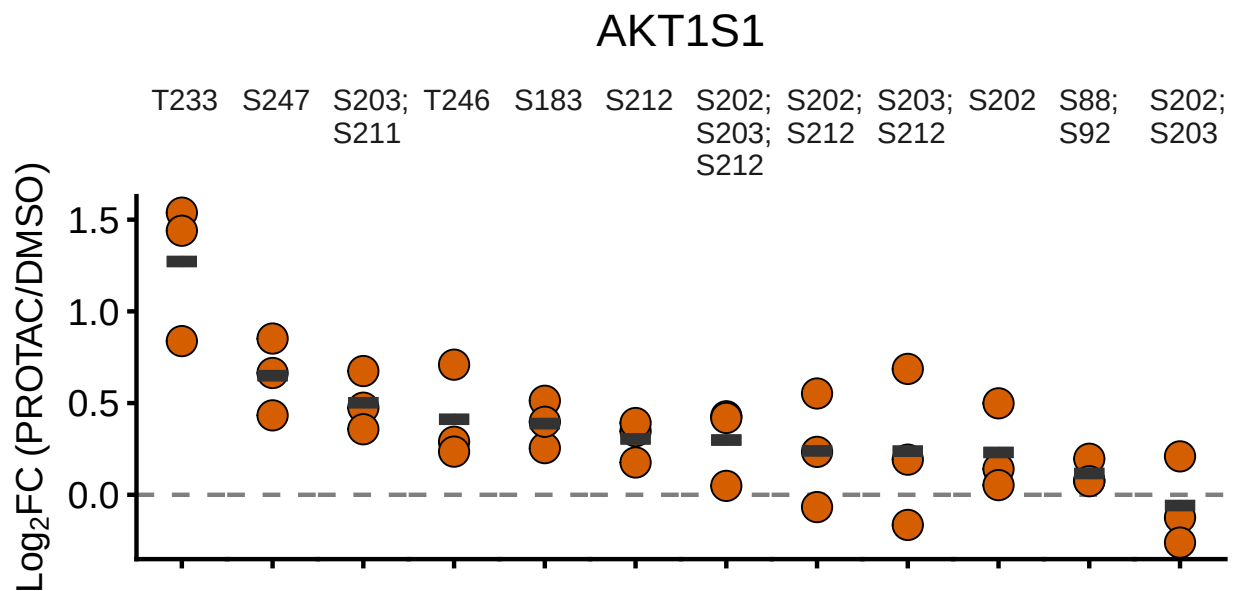
```
ggsave(plot=p1, here('results/plots/PROTAC_AKT1S1_phosphosite_changes_selected.png'))
```

Saving 6.5 x 4.5 in image

```
ggsave(plot=p1, here('results/plots/PROTAC_AKT1S1_phosphosite_changes_selected.pdf'))
```

Saving 6.5 x 4.5 in image

```
p2 <- p + filter(to_plot, condition!='DMSO')
print(p2)
```



```
ggsave(plot=p2, here('results/plots/PROTAC_AKT1S1_phosphosite_changes.png'))
```

```
## Saving 6.5 x 4.5 in image
```

```
ggsave(plot=p2, here('results/plots/PROTAC_AKT1S1_phosphosite_changes.pdf'))
```

```
## Saving 6.5 x 4.5 in image
```

```
obj <- qfs$total
poi <- 'Q96B36'
experiments_to_plot = c('psm_raw', 'psm_filtered_forSum')
protein_id_col = "Leading.razor.protein"
label_col = "Gene.Names.First"
log2transform_cols = c('psm_raw', 'psm_filtered_forSum')
alpha_assays = FALSE
row_vars <- 'Sequence'
```

```
rowData(qfs$total[['psm_raw']]) %>%
  data.frame() %>%
  filter(Leading.razor.protein %in% poi)
```

```
##      Leading.razor.protein Sequence Length.x Modifications Modified.sequence
##      Oxidation..M..Probabilities Phospho..STY..Probabilities
```

```

##      Oxidation..M..Score.Diffs Phospho..STY..Score.Diffs
##      Acetyl..Protein.N.term. Oxidation..M. Phospho..STY. Missed.cleavages
##      Proteins Leading.proteins Gene.names Protein.names.x Type Raw.file
##      MS.MS.m.z Charge m.z Mass Uncalibrated...Calibrated.m.z..ppm.
##      Uncalibrated...Calibrated.m.z..Da. Mass.error..ppm. Mass.error..Da.
##      Uncalibrated.mass.error..ppm. Uncalibrated.mass.error..Da.
##      Max.intensity.m.z.0 Retention.time Retention.length
##      Calibrated.retention.time Calibrated.retention.time.start
##      Calibrated.retention.time.finish Retention.time.calibration
##      Match.time.difference Match.m.z.difference Match.q.value Match.score
##      Number.of.data.points Number.of.scans Number.of.isotopic.peaks PIF
##      Fraction.of.total.spectrum Base.peak.fraction PEP MS.MS.count
##      MS.MS.scan.number MS.MS.scan.numbers MS3.scan.numbers Score Delta.score
##      Combinatorics Intensity Reporter.intensity.1 Reporter.intensity.2
##      Reporter.intensity.3 Reporter.intensity.4 Reporter.intensity.5
##      Reporter.intensity.6 Reporter.intensity.7 Reporter.intensity.8
##      Reporter.intensity.9 Reporter.intensity.10 Reporter.intensity.11
##      Reporter.intensity.12 Reporter.intensity.13 Reporter.intensity.14
##      Reporter.intensity.15 Reporter.intensity.16 Reporter.intensity.17
##      Reporter.intensity.18 Reporter.intensity.count.1
##      Reporter.intensity.count.2 Reporter.intensity.count.3
##      Reporter.intensity.count.4 Reporter.intensity.count.5
##      Reporter.intensity.count.6 Reporter.intensity.count.7
##      Reporter.intensity.count.8 Reporter.intensity.count.9
##      Reporter.intensity.count.10 Reporter.intensity.count.11
##      Reporter.intensity.count.12 Reporter.intensity.count.13
##      Reporter.intensity.count.14 Reporter.intensity.count.15
##      Reporter.intensity.count.16 Reporter.intensity.count.17
##      Reporter.intensity.count.18 Reverse Potential.contaminant id
##      Protein.group.IDs Peptide.ID Mod..peptide.ID MS.MS.IDs Best.MS.MS
##      Oxidation..M..site.IDs
##      [ reached 'max' / getOption("max.print") -- omitted 12 columns ]
##      [ reached 'max' / getOption("max.print") -- omitted 7 rows ]

```

```

to_plot_pep <- obj[, , experiments_to_plot] %>% biomasslmb:::qfeatures_long(rowvars = c(protein_id_col,
  label_col, row_vars)) %>% data.frame()

```

```

## Warning: 'experiments' dropped; see 'drops()'

```

```

## harmonizing input:
## removing 24 sampleMap rows not in names(experiments)

```

```

to_plot_prot <- obj[, , 'protein'] %>% biomasslmb:::qfeatures_long(rowvars = c(protein_id_col,
  label_col)) %>% data.frame()

```

```

## Warning: 'experiments' dropped; see 'drops()'

```

```

## harmonizing input:
## removing 30 sampleMap rows not in names(experiments)

```

```
to_plot <- bind_rows(to_plot_prot, to_plot_pep) %>%
  filter(!is.na(protein_id_col) %in% poi) %>%
  mutate(label = !is.na(label_col)) %>%
  mutate(value = ifelse(assay %in% log2transform_cols, log2(value), value)) %>%
  mutate(assay_order = factor(assay, levels = c(experiments_to_plot, 'protein')))
```

```
to_plot <- to_plot %>%
  merge(colData(obj), by.x='colname', by.y='row.names')
```

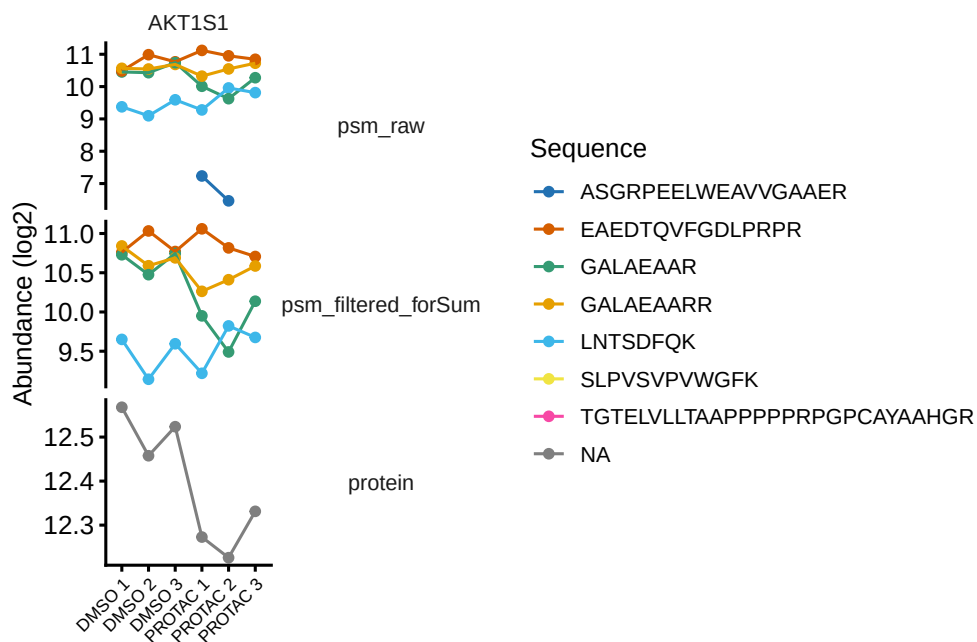
```
ylab <- "Abundance (log2)"
```

```
p <- to_plot %>%
  ggplot(aes(x = paste(condition, replicate), y = value)) + geom_point() +
  geom_line(aes(group = rowname)) + facet_grid(assay_order ~
    label, scales = "free_y") + theme_biomasslmb(border = FALSE,
    base_size = 12) + theme(axis.text.x = element_text(angle = 45,
    vjust = 1, hjust = 1, size = 8), strip.text.y = element_text(size = 10,
    angle = 0), strip.background = element_blank(), panel.spacing.x = unit(10,
    "mm")) + labs(x = "", y = ylab) +
  aes(colour=Sequence) +
  scale_colour_manual(values=get_cat_palette(7))
```

p

```
## Warning: Removed 16 rows containing missing values or values outside the scale range
## ('geom_point()').
```

```
## Warning: Removed 16 rows containing missing values or values outside the scale range
## ('geom_line()').
```



```
mtor_substrates <- read.csv(here('raw/PROTAC/mTORC1_substrates_table1_expanded.csv'))
```

```
mtor_substrates
```

```
##      Substrate Uniprot.ID Phosphorylated.site.s.
## 1      4E-BP1      Q13541      T37
## 2      4E-BP1      Q13541      T46
## 3      4E-BP1      Q13541      S65
## 4      4E-BP1      Q13541      T70
## 5      ACINUS      Q9UKV3      S240
## 6      ACINUS      Q9UKV3      S243
## 7      AMBRA1      Q9COC7      S52
## 8      AMPKa1      Q13131      S356
## 9      AMPKa2      P54646      S345
## 10     AMPKa2      P54646      S377
## 11      AR        P10275      S96
## 12     AS160      O60343      S666
##      Phosphorylated.site.s..and.adjacent.residues TOS.motif
## 1      GDYSTTPGGTLFSTTPGGTR      FEMDI
## 2      GDYSTTPGGTLFSTTPGGTR      FEMDI
## 3      MECRNSPVTKTTPRDL
## 4      MECRNSPVTKTTPRDL
## 5      RAAKLSEGSQPAEE
## 6      RAAKLSEGSQPAEE
## 7      VELPDSPRSTF
## 8      FYLATSPDPSF
## 9      FYLASSPPSGS
## 10     PLIADSPKARC
## 11     QGEDGSPQAGR
## 12     AQGVRSPLLRQ      FEMDI
##      Purpose.of.phosphorylation      Function
## 1      Inhibitory Translation regulation
## 2      Inhibitory Translation regulation
## 3      Inhibitory Translation regulation
## 4      Inhibitory Translation regulation
## 5      Unknown significance      EJC/splicing
## 6      Unknown significance      EJC/splicing
## 7      Inhibitory      Autophagy
## 8      Inhibitory      Signaling
## 9      Inhibitory      Signaling
## 10     Specific activation      Glucose uptake
## 11     Activating      Signaling
## 12     Unknown      Signaling
##      Remarks
## 1      Rapamycin insensitive
## 2      Rapamycin insensitive
## 3      Rapamycin sensitive
## 4      Rapamycin sensitive
## 5
## 6
## 7
## 8
```

```
## 9
## 10
## 11
## 12 Sensitive to TOS mutation and rapamycin
## [ reached 'max' / getOption("max.print") -- omitted 95 rows ]
```

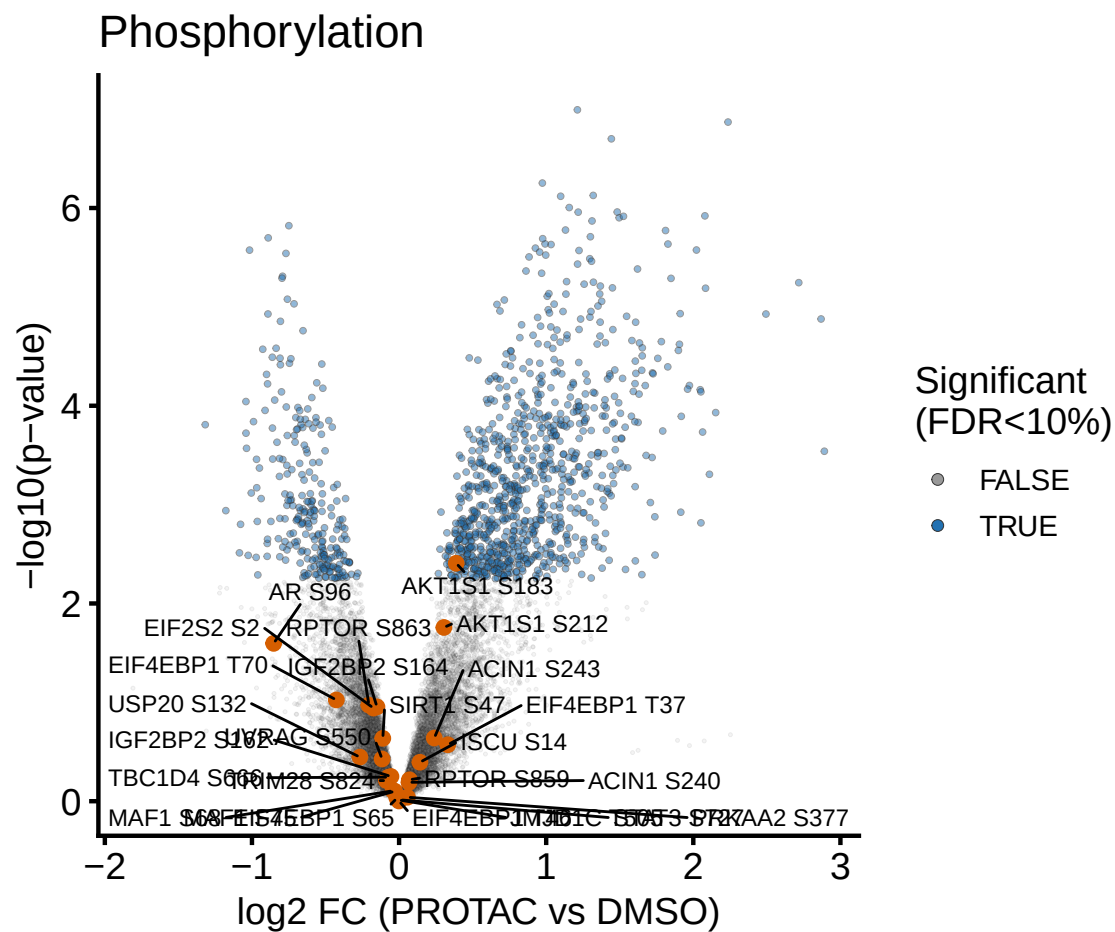
```
# highlight
highlight <- phospho_limma_results %>%
  merge(mtor_substrates,
        by.x=c('UniprotID', 'phospho_pos'),
        by.y=c('Uniprot.ID', 'Phosphorylated.site.s.'))

print(nrow(highlight))
```

```
## [1] 25
```

```
phospho_limma_results %>%
  mutate(significant=adj.P.Val<0.1) %>%
  ggplot(aes(logFC, -log10(P.Value))) +
  geom_point(aes(fill=significant,
                 alpha=significant, size=significant),
             colour='grey20', stroke=0.25, shape=21) +
  geom_point(data=highlight, colour=get_cat_palette(2)[2]) +
  theme_biomasslmb() +
  scale_fill_manual(values=c('grey60', get_cat_palette(1)),
                    name='Significant\n(FDR<10%)') +
  scale_size_manual(values=c(0.5, 1), guide='none') +
  scale_alpha_manual(values=c(0.1, 0.5), guide='none') +
  labs(x='log2 FC (PROTAC vs DMSO)', y='-log10(p-value)') +
  theme(strip.background = element_blank()) +
  geom_text_repel(
    data=highlight,
    aes(label=paste(Gene.Names.First, gsub('.*_ ', '', phosphosite))),
    max.overlaps = 50, size=3, min.segment.length =0, colour='black', force = 10) +
  ggtitle('Phosphorylation') -> p

print(p)
```



mtor_substrates

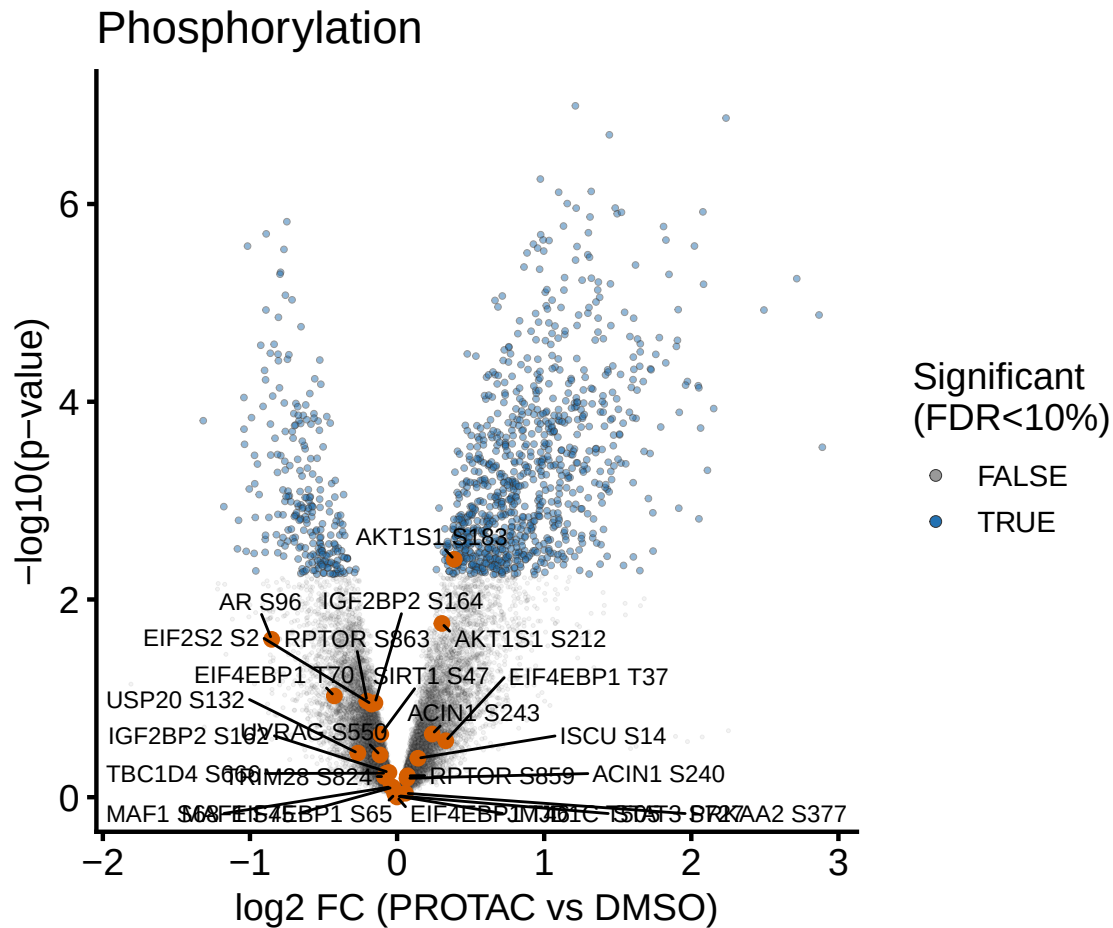
##	Substrate	Uniprot.ID	Phosphorylated.site.s.	
## 1	4E-BP1	Q13541	T37	
## 2	4E-BP1	Q13541	T46	
## 3	4E-BP1	Q13541	S65	
## 4	4E-BP1	Q13541	T70	
## 5	ACINUS	Q9UKV3	S240	
## 6	ACINUS	Q9UKV3	S243	
## 7	AMBRA1	Q9COC7	S52	
## 8	AMPKa1	Q13131	S356	
## 9	AMPKa2	P54646	S345	
## 10	AMPKa2	P54646	S377	
## 11	AR	P10275	S96	
## 12	AS160	O60343	S666	
##	Phosphorylated.site.s..and.adjacent.residues			TOS.motif
## 1		GDYSTTPGGTLFSTTPGGTR		FEMDI


```

## 2          GDYSTTPGGTLFSTTPGGTR      FEMDI
## 3          MECRNSPVTKTPPRDL
## 4          MECRNSPVTKTPPRDL
## 5          RAAKLSEGSQPAEE
## 6          RAAKLSEGSQPAEE
## 7          VELPDSPRSTF
## 8          FYLATSPDPSF
## 9          FYLASSPPSGS
## 10         PLIADSPKARC
## 11         QGEDGSPQHR
## 12         AQGVRSPLLRQ      FEMDI
## Purpose.of.phosphorylation      Function
## 1          Inhibitory Translation regulation
## 2          Inhibitory Translation regulation
## 3          Inhibitory Translation regulation
## 4          Inhibitory Translation regulation
## 5          Unknown significance      EJC/splicing
## 6          Unknown significance      EJC/splicing
## 7          Inhibitory      Autophagy
## 8          Inhibitory      Signaling
## 9          Inhibitory      Signaling
## 10         Specific activation      Glucose uptake
## 11         Activating      Signaling
## 12         Unknown      Signaling
##          Remarks
## 1          Rapamycin insensitive
## 2          Rapamycin insensitive
## 3          Rapamycin sensitive
## 4          Rapamycin sensitive
## 5
## 6
## 7
## 8
## 9
## 10
## 11
## 12 Sensitive to TOS mutation and rapamycin
## [ reached 'max' / getOption("max.print") -- omitted 95 rows ]

```

```
print(p)
```



```
# highlight
highlight2 <- phospho_limma_results %>%
  filter(UniprotID %in% unique(mtor_substrates$Uniprot.ID))

mtor_substrates %>%
  filter(Uniprot.ID %in% pull(filter(highlight2, adj.P.Val<0.1), UniprotID)) %>%
  select(Substrate, Uniprot.ID, Phosphorylated.site.s., Function, Remarks) %>%
  merge(distinct(select(highlight, Gene.Names.First, Gene.Names, UniprotID, Protein.names.y)),
    by.x='Uniprot.ID', by.y='UniprotID')
```

##	Uniprot.ID	Substrate	Phosphorylated.site.s.	Function
## 1	Q15652	JMJD1C	T505	Histone demethylase
## 2	Q96B36	PRAS40	S183	Signaling
## 3	Q96B36	PRAS40	S212	Signaling
## 4	Q96B36	PRAS40	S221	Signaling

```

## 5      Q9UKV3      ACINUS      S240      EJC/splicing
## 6      Q9UKV3      ACINUS      S243      EJC/splicing
##
##      Remarks Gene.Names.First
## 1
## 2      mTORC1 regulator      AKT1S1
## 3 S212 rapamycin insensitive      AKT1S1
## 4 S212 rapamycin insensitive      AKT1S1
## 5
## 6      ACIN1
## 6      ACIN1
##
##      Gene.Names
## 1 JMJD1C JHDM2C KDM3C KIAA1380 TRIP8
## 2      AKT1S1 PRAS40
## 3      AKT1S1 PRAS40
## 4      AKT1S1 PRAS40
## 5      ACIN1 ACINUS KIAA0670
## 6      ACIN1 ACINUS KIAA0670
##
## 1 Jumonji domain-containing protein 1C (EC 1.14.11.-) (Thyroid receptor-interacting protein 8) (TR-i
## 2      Proline-rich AKT1 substrate 1 (40 k
## 3      Proline-rich AKT1 substrate 1 (40 k
## 4      Proline-rich AKT1 substrate 1 (40 k
## 5      Apoptotic chromatin condensation i
## 6      Apoptotic chromatin condensation i

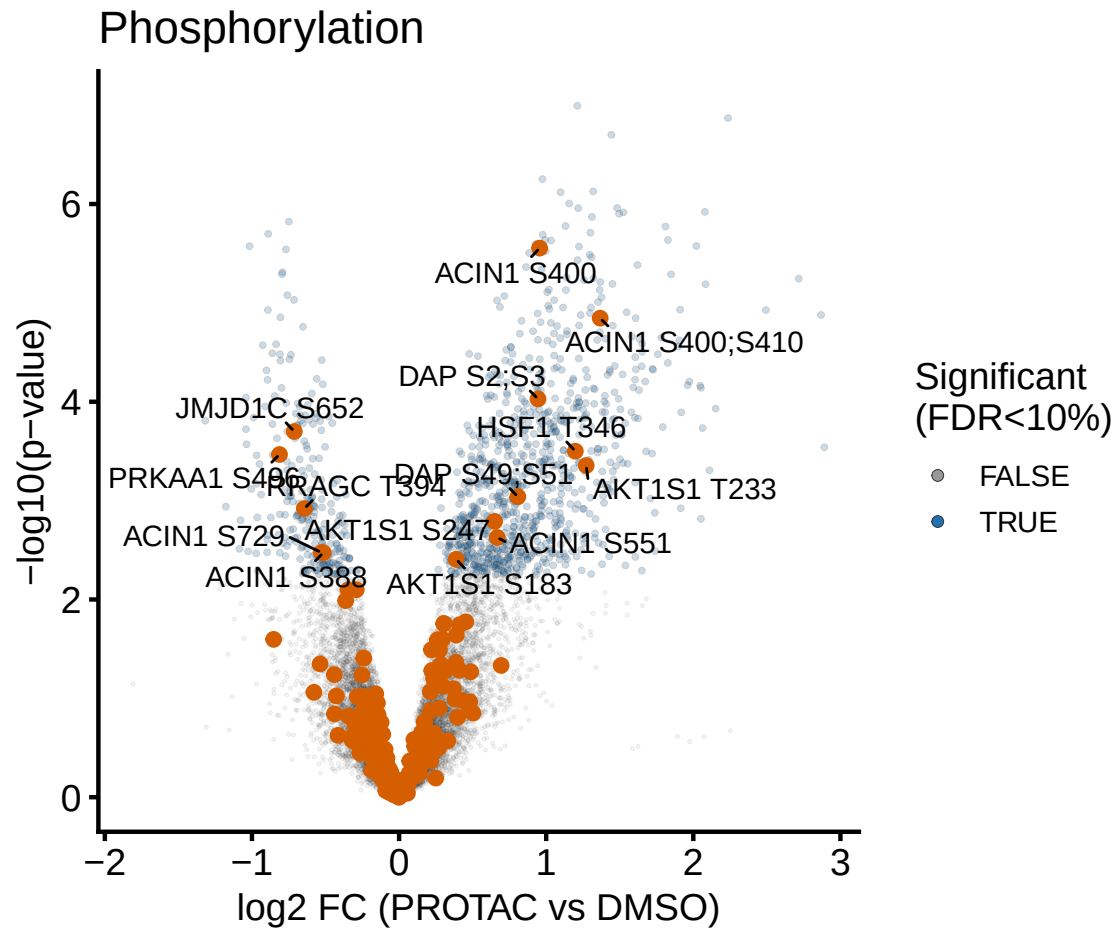
```

```

phospho_limma_results %>%
  mutate(significant=adj.P.Val<0.1) %>%
  ggplot(aes(logFC, -log10(P.Value))) +
  geom_point(aes(fill=significant,
    alpha=significant, size=significant),
    colour='grey20', stroke=0.25, shape=21) +
  geom_point(data=highlight2, colour=get_cat_palette(2)[2]) +
  theme_biomasslmb() +
  scale_fill_manual(values=c('grey60', get_cat_palette(1)),
    name='Significant\n(FDR<10%)') +
  scale_size_manual(values=c(0.5, 1), guide='none') +
  scale_alpha_manual(values=c(0.1, 0.25), guide='none') +
  labs(x='log2 FC (PROTAC vs DMSO)', y='-log10(p-value)') +
  theme(strip.background = element_blank()) +
  geom_text_repel(
    data=filter(highlight2, adj.P.Val<0.1),
    aes(label=paste(Gene.Names.First, gsub('.*_', '', phosphosite))),
    max.overlaps = 50, size=4, min.segment.length =0, colour='black', force = 10) +
  ggtitle('Phosphorylation') -> p

print(p)

```



```

crosstab_poi_sig <- phospho_limma_results %>%
  group_by(is_poi=ifelse(phosphosite %in% highlight2$phosphosite, 'mTORC1 substrate', 'Other'),
    significant=adj.P.Val<0.1) %>%
  tally() %>%
  pivot_wider(names_from = significant, values_from=n, values_fill=0) %>%
  tibble::column_to_rowname('is_poi')

crosstab_poi_sig %>%
  mutate(frac_sig=`TRUE`/(`TRUE`+`FALSE`))

```

```

##           FALSE TRUE  frac_sig
## Other      18746 1113 0.05604512
## mTORC1 substrate    223   14 0.05907173

```

```
print(crosstab_poi_sig)
```

```
##                FALSE TRUE
## Other          18746 1113
## mTORC1 substrate   223   14
```

```
fisher.test(crosstab_poi_sig)
```

```
##
## Fisher's Exact Test for Count Data
##
## data:  crosstab_poi_sig
## p-value = 0.7765
## alternative hypothesis: true odds ratio is not equal to 1
## 95 percent confidence interval:
##  0.5668772 1.8197239
## sample estimates:
## odds ratio
##  1.057415
```

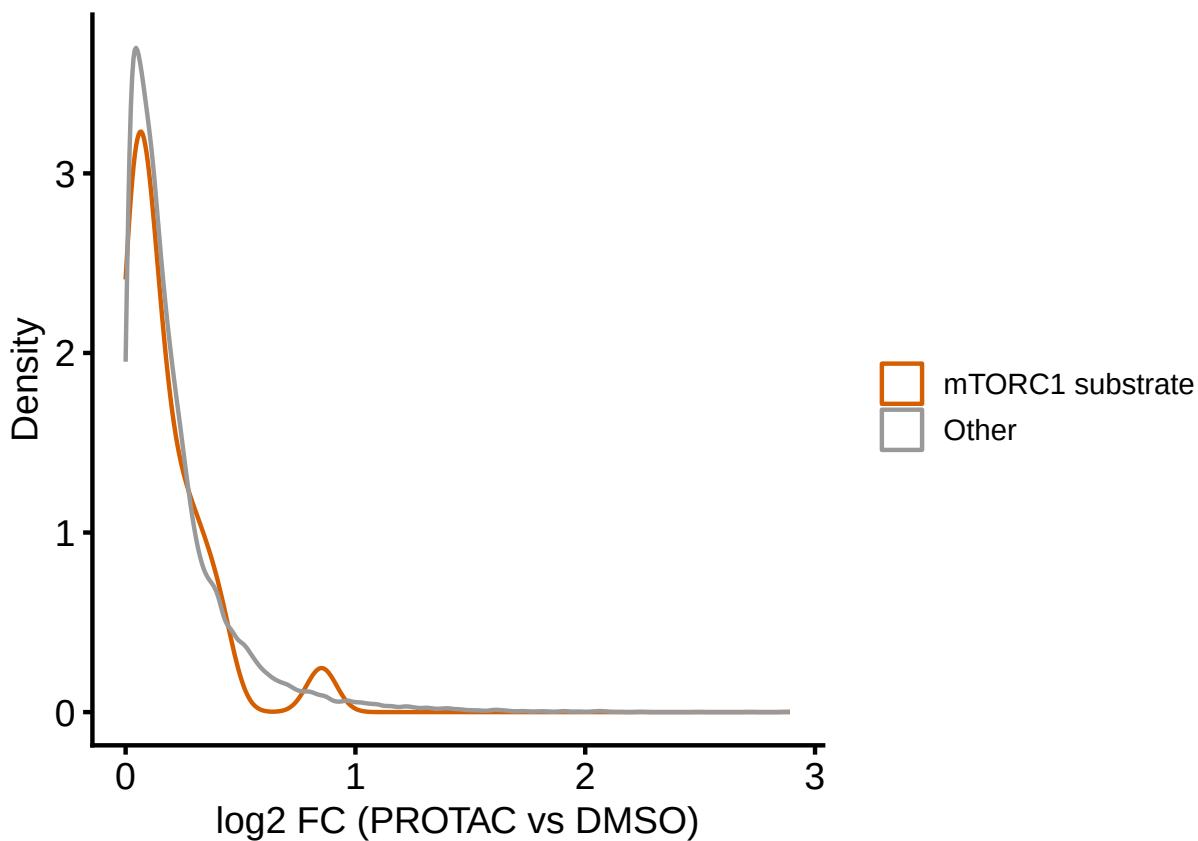
```
phospho_limma_results %>%
```

```
  mutate(is_poi = ifelse(UniprotID %in% unique(mtor_substrates$Uniprot.ID), 'mTORC1 substrate', 'Other')) %>%
  dplyr::count(is_poi)
```

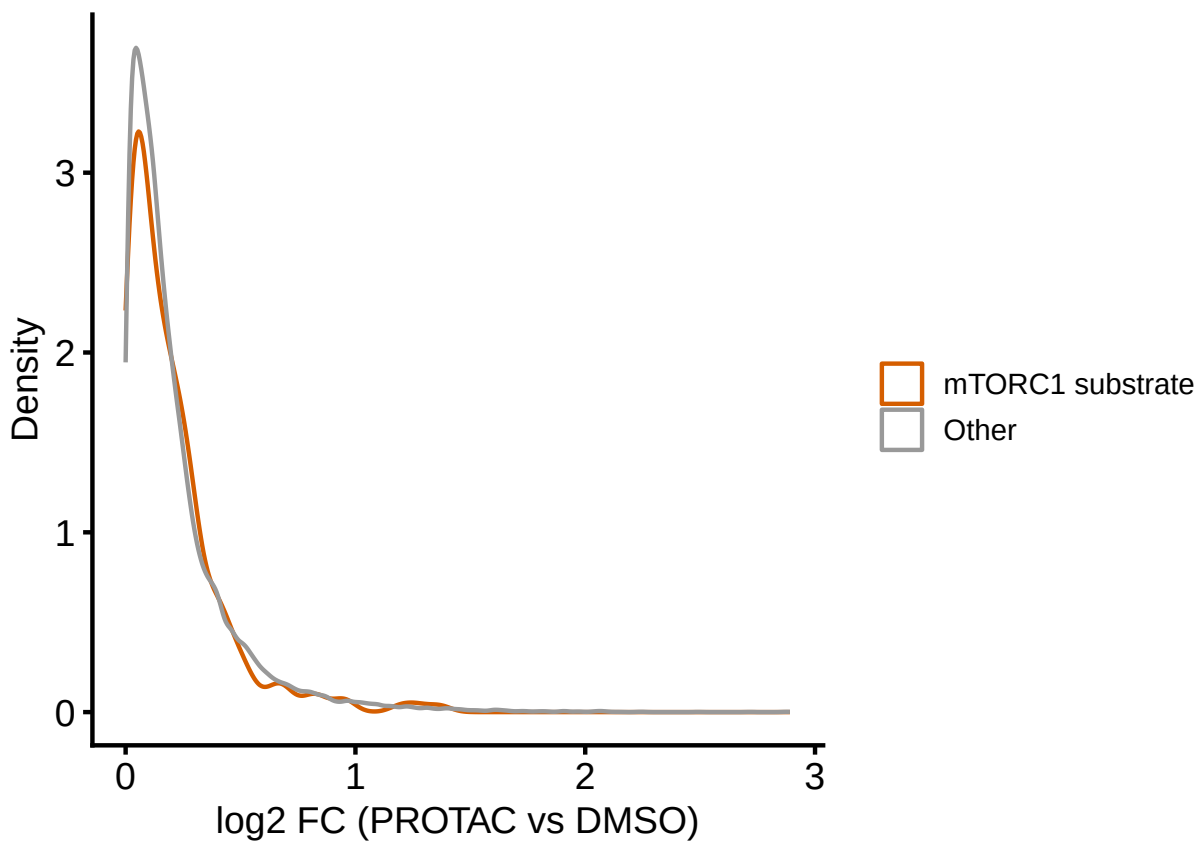
```
##          is_poi      n
## 1             Other 19859
## 2 mTORC1 substrate   237
```

```
phospho_limma_results %>%
```

```
  mutate(is_poi = ifelse(phosphosite %in% unique(highlight$phosphosite), 'mTORC1 substrate', 'Other')) %>%
  ggplot(aes(abs(logFC), colour=is_poi)) +
  geom_density() +
  theme_biomasslmb() +
  labs(x='log2 FC (PROTAC vs DMSO)', y='Density', colour='') +
  scale_colour_manual(values=c(get_cat_palette(2)[2], 'grey60'))
```



```
phospho_limma_results %>%
  mutate(is_poi = ifelse(UniprotID %in% unique(mtor_substrates$Uniprot.ID), 'mTORC1 substrate', 'Other'))
ggplot(aes(abs(logFC), colour=is_poi)) +
  geom_density() +
  theme_biomasslmb() +
  labs(x='log2 FC (PROTAC vs DMSO)', y='Density', colour='') +
  scale_colour_manual(values=c(get_cat_palette(2)[2], 'grey60'))
```



```
phospho_limma_results %>%
  mutate(is_poi = ifelse(phosphosite %in% unique(highlight$phosphosite), 'mTORC1 substrate', 'Other')) %>%
  ks.test(abs(logFC) ~ is_poi, data=.)
```

```
##
## Asymptotic two-sample Kolmogorov-Smirnov test
##
## data: abs(logFC) by is_poi
## D = 0.16528, p-value = 0.5027
## alternative hypothesis: two-sided
```

```
phospho_limma_results %>%
  mutate(is_poi = ifelse(UniprotID %in% unique(mtor_substrates$Uniprot.ID), 'mTORC1 substrate', 'Other')) %>%
  ks.test(abs(logFC) ~ is_poi, data=.)
```

```
##
## Asymptotic two-sample Kolmogorov-Smirnov test
##
## data: abs(logFC) by is_poi
## D = 0.065492, p-value = 0.2676
## alternative hypothesis: two-sided
```

```
set.seed(0)
```

```
mtor_substrates_updated <- openxlsx::read.xlsx(here('raw/PROTAC/mTORC1_substrates_Battaglioni2022_plus_1'),
data.frame())
```

```
nrow(mtor_substrates)
```

```
## [1] 107
```

```
nrow(mtor_substrates_updated)
```

```
## [1] 136
```

```
merge(mtor_substrates, mtor_substrates_updated, by=c('Uniprot.ID', 'Phosphorylated.site.s.')) %>%
  nrow()
```

```
## [1] 107
```

```
phospho_limma_results %>%
  merge(mtor_substrates_updated,
        by.x=c('UniprotID', 'phospho_pos'),
        by.y=c('Uniprot.ID', 'Phosphorylated.site.s.')) %>%
  select(Gene.Names.First, Entry.Name)
```

```
##      Gene.Names.First  Entry.Name
## 1          PRKAB2  AAKB2_HUMAN
## 2          TBC1D4  TBCD4_HUMAN
## 3              AR   ANDR_HUMAN
## 4          EIF2S2  IF2B_HUMAN
## 5          STAT3  STAT3_HUMAN
## 6          PRKAA2  AAPK2_HUMAN
## 7          TRIM28  TIF1B_HUMAN
## 8      EIF4EBP1  4EBP1_HUMAN
## 9      EIF4EBP1  4EBP1_HUMAN
## 10     EIF4EBP1  4EBP1_HUMAN
## 11     EIF4EBP1  4EBP1_HUMAN
## 12          JMD1C  JHD2C_HUMAN
## 13          RPTOR  RPTOR_HUMAN
## 14          RPTOR  RPTOR_HUMAN
## 15          SERBP1  SERB1_HUMAN
## 16          AKT1S1  AKTS1_HUMAN
## 17          AKT1S1  AKTS1_HUMAN
## 18          SIRT1  SIR1_HUMAN
## 19          MAF1  MAF1_HUMAN
## 20          MAF1  MAF1_HUMAN
## 21          ISCU  ISCU_HUMAN
## 22          UVRAG  UVRAG_HUMAN
## 23          ACIN1  ACINU_HUMAN
## 24          ACIN1  ACINU_HUMAN
## 25          USP20  UBP20_HUMAN
## 26      IGF2BP2  IF2B2_HUMAN
## 27      IGF2BP2  IF2B2_HUMAN
```



```

# highlight
highlight_val <- phospho_limma_results %>%
  merge(mtor_substrates_updated,
        by.x=c('UniprotID', 'phospho_pos'),
        by.y=c('Uniprot.ID', 'Phosphorylated.site.s.')) %>%
  select(logFC, P.Value, adj.P.Val, Gene.Names.First, Entry.Name, phosphosite) %>%
  mutate(type='in vitro validated')

highlight_val2 <- phospho_limma_results %>%
  filter(phosphosite %in% c('P51397_S2;S3', 'P51397_S49;S51')) %>%
  select(logFC, P.Value, adj.P.Val, Gene.Names.First, Entry.Name, phosphosite) %>%
  mutate(type='in vitro validated')

highlight_val <- bind_rows(highlight_val, highlight_val2) %>%
  distinct()

highlight_all_sites_in_protein <- phospho_limma_results %>%
  filter(UniprotID %in% unique(mtor_substrates$Uniprot.ID)) %>%
  filter(!phosphosite %in% highlight_val$phosphosite) %>%
  select(logFC, P.Value, adj.P.Val, Gene.Names.First, Entry.Name, phosphosite) %>%
  mutate(type='other site on validated protein')

highlight <- bind_rows(highlight_val, highlight_all_sites_in_protein) %>%
  mutate(significant=adj.P.Val<0.1)

highlight %>%
  filter(Gene.Names.First=='AKT1S1')

```

##	logFC	P.Value	adj.P.Val	Gene.Names.First	Entry.Name
## 1	0.38828008	0.0039193037	0.08032776	AKT1S1	AKTS1_HUMAN
## 2	0.30470177	0.0174740857	0.19311543	AKT1S1	AKTS1_HUMAN
## 3	0.23077510	0.2762929065	0.68868626	AKT1S1	AKTS1_HUMAN
## 4	-0.05863947	0.7837159438	0.94237166	AKT1S1	AKTS1_HUMAN
## 5	0.29842627	0.0753778107	0.41417673	AKT1S1	AKTS1_HUMAN
## 6	0.23970113	0.2884430766	0.69895130	AKT1S1	AKTS1_HUMAN
## 7	0.50224098	0.1410958315	0.53763023	AKT1S1	AKTS1_HUMAN
## 8	0.23772754	0.3082356346	0.71544751	AKT1S1	AKTS1_HUMAN
## 9	0.64961021	0.0016238829	0.04709026	AKT1S1	AKTS1_HUMAN
## 10	0.11611963	0.4860088632	0.82129449	AKT1S1	AKTS1_HUMAN
## 11	1.27191851	0.0004413551	0.02088897	AKT1S1	AKTS1_HUMAN
## 12	0.41125695	0.0180523901	0.19535855	AKT1S1	AKTS1_HUMAN

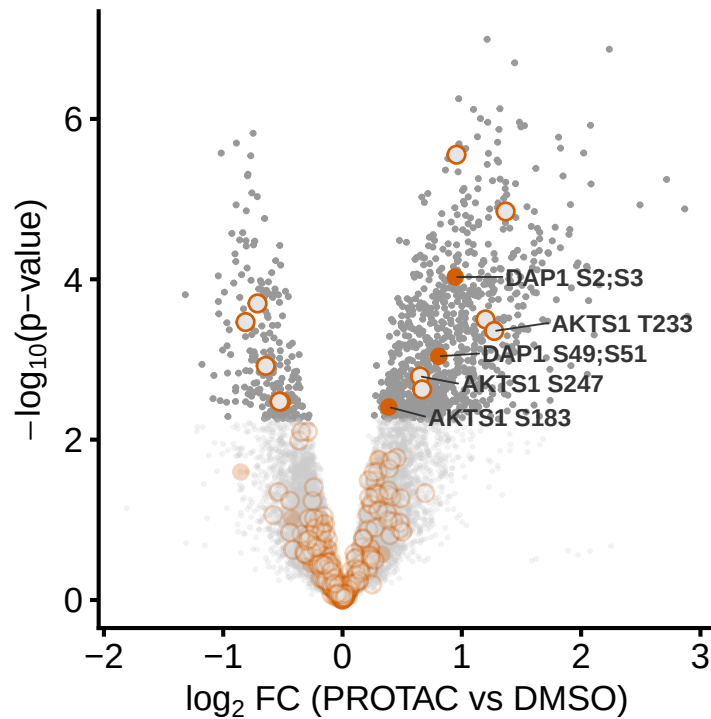
##	phosphosite	type	significant
## 1	Q96B36_S183	in vitro validated	TRUE
## 2	Q96B36_S212	in vitro validated	FALSE
## 3	Q96B36_S202	other site on validated protein	FALSE
## 4	Q96B36_S202;S203	other site on validated protein	FALSE
## 5	Q96B36_S202;S203;S212	other site on validated protein	FALSE
## 6	Q96B36_S202;S212	other site on validated protein	FALSE
## 7	Q96B36_S203;S211	other site on validated protein	FALSE
## 8	Q96B36_S203;S212	other site on validated protein	FALSE
## 9	Q96B36_S247	other site on validated protein	TRUE
## 10	Q96B36_S88;S92	other site on validated protein	FALSE

```
## 11          Q96B36_T233 other site on validated protein      TRUE
## 12          Q96B36_T246 other site on validated protein      FALSE
```

```
phospho_limma_results %>%
  filter(adj.P.Val<0.1) %>%
  ggplot(aes(logFC, -log10(P.Value))) +
  geom_point(colour='grey60', alpha=1, size=1, shape=20) +
  geom_point(data = filter(phospho_limma_results, adj.P.Val>0.1),
    colour='grey80', alpha=0.25, size=0.5, shape=20) +

  geom_point(data=filter(highlight, type=='in vitro validated'), colour=get_cat_palette(2)[2],
    aes(alpha=significant), pch=16, size=4) +
  geom_point(data=filter(highlight, type=='other site on validated protein'),
    colour=get_cat_palette(2)[2],
    fill='grey90',
    aes(alpha=significant), pch=21, size=4) +
  theme_biomasslmb(base_size = 20) +
  scale_fill_manual(values=c(get_cat_palette(2)[2], 'grey90'), name='mTORC1 substrate') +
  scale_size_manual(values=c(0.5, 1), guide='none') +
  scale_alpha_manual(values=c(0.25, 1), guide='none') +
  labs(x = expression(log[2]*" FC (PROTAC vs DMSO)"),
    y = expression(-log[10]*"(p-value)")) +
  theme(strip.background = element_blank()) +
  geom_text_repel(
    data=filter(highlight, Gene.Names.First == 'DAP'|phosphosite %in% c('Q96B36_S183', 'Q96B36_T233', 'Q96B36_T246')),
    aes(label=paste(sub('_HUMAN', '', Entry.Name), gsub('.*_', '', phosphosite))),
    fontface = "bold",
    max.overlaps = 50, size=5, min.segment.length = 0, colour='grey20', force = 20, nudge_x = 1) -> p

print(p)
```



```
ggsave(plot=p, here('results/plots/PROTAC_phospho_limma_results_mTORC1_substrates.png'))
```

```
## Saving 10 x 6 in image
```

```
ggsave(plot=p, here('results/plots/PROTAC_phospho_limma_results_mTORC1_substrates.pdf'))
```

```
## Saving 10 x 6 in image
```

```
crosstab_poi_sig <- phospho_limma_results %>%
  group_by(is_poi=ifelse(phosphosite %in% highlight_val$phosphosite, 'mTORC1 substrate', 'Other'),
            significant=adj.P.Val<0.1) %>%
  tally() %>%
  pivot_wider(names_from = significant, values_from=n, values_fill=0) %>%
  tibble::column_to_rowname('is_poi')

crosstab_poi_sig %>%
  mutate(frac_sig=`TRUE`/(`TRUE`+`FALSE`))
```

```
##           FALSE TRUE  frac_sig
## Other      18943 1124 0.05601236
## mTORC1 substrate    26    3 0.10344828
```

```
print(crosstab_poi_sig)
```

```
##           FALSE TRUE
## Other      18943 1124
## mTORC1 substrate    26    3
```

```
fisher.test(crosstab_poi_sig)
```

```
##
## Fisher's Exact Test for Count Data
##
## data:  crosstab_poi_sig
## p-value = 0.2205
## alternative hypothesis: true odds ratio is not equal to 1
## 95 percent confidence interval:
##  0.3761309 6.3605363
## sample estimates:
## odds ratio
##  1.944541
```

```
crosstab_poi_sig <- phospho_limma_results %>%
  group_by(is_poi=ifelse(phosphosite %in% highlight$phosphosite, 'mTORC1 substrate', 'Other'),
            significant=adj.P.Val<0.1) %>%
  tally() %>%
  pivot_wider(names_from = significant, values_from=n, values_fill=0) %>%
  tibble::column_to_rownames('is_poi')

crosstab_poi_sig %>%
  mutate(frac_sig=`TRUE`/(`TRUE`+`FALSE`))
```

```
##              FALSE TRUE  frac_sig
## Other          18744 1113 0.05605076
## mTORC1 substrate   225   14 0.05857741
```

```
print(crosstab_poi_sig)
```

```
##              FALSE TRUE
## Other          18744 1113
## mTORC1 substrate   225   14
```

```
fisher.test(crosstab_poi_sig)
```

```
##
## Fisher's Exact Test for Count Data
##
## data:  crosstab_poi_sig
## p-value = 0.7783
## alternative hypothesis: true odds ratio is not equal to 1
## 95 percent confidence interval:
##  0.5618574 1.8030059
## sample estimates:
## odds ratio
##  1.047898
```

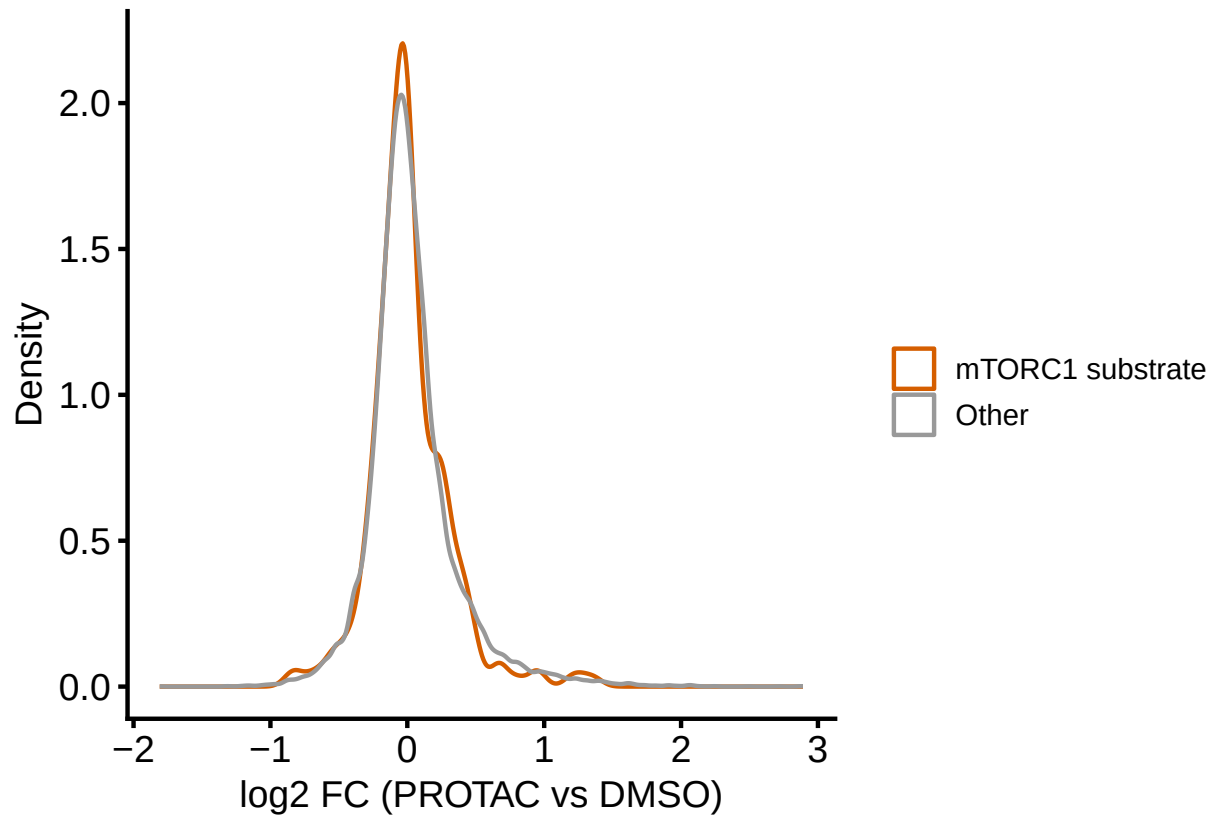
```
rowData(qfs$phospho[['psm_annot']]) %>%
  data.frame() %>%
  filter(Gene.Names.First=='DAP') %>%
  select(Entry.Name, ptm_name, Phospho..STY..Probabilities) %>%
  knitr::kable()
```

	Entry.Name	ptm_name	Phospho..STY..Probabilities
19053	DAP1_HUMANS49; S51		DKDDQEWES(1)PS(0.998)PPKPT(0.002)VFISGVIAIR
19055	DAP1_HUMANS49; S51		DKDDQEWES(0.994)PS(0.98)PPKPT(0.024)VFIS(0.001)GVIAIR
19056	DAP1_HUMANS2; S3		S(1)S(1)PPEGKLETK

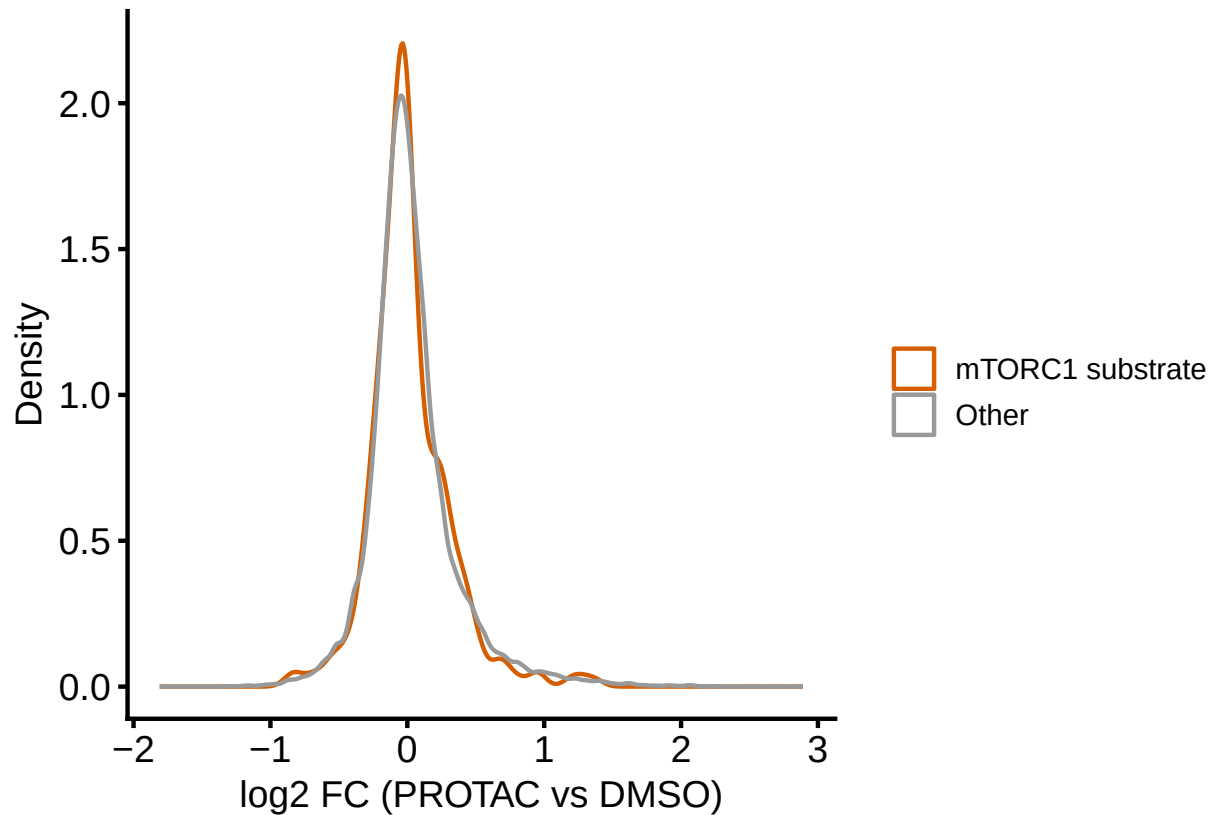
```
phospho_limma_results %>%
  mutate(is_poi = ifelse(UniprotID %in% unique(mtor_substrates_updated$Uniprot.ID), 'mTORC1 substrate',
    count(is_poi)
```

```
##           is_poi      n
## 1           Other 19827
## 2 mTORC1 substrate   269
```

```
phospho_limma_results %>%
  mutate(is_poi = ifelse(phosphosite %in% unique(highlight$phosphosite), 'mTORC1 substrate', 'Other')) %>%
  ggplot(aes(logFC, colour=is_poi)) +
  geom_density() +
  theme_biomasslmb() +
  labs(x='log2 FC (PROTAC vs DMSO)', y='Density', colour='') +
  scale_colour_manual(values=c(get_cat_palette(2)[2], 'grey60'))
```



```
phospho_limma_results %>%
  mutate(is_poi = ifelse(UniprotID %in% unique(mtor_substrates_updated$Uniprot.ID), 'mTORC1 substrate',
    ggplot(aes(logFC, colour=is_poi)) +
    geom_density() +
    theme_biomasslmb() +
    labs(x='log2 FC (PROTAC vs DMSO)', y='Density', colour='') +
    scale_colour_manual(values=c(get_cat_palette(2)[2], 'grey60'))
```



```
phospho_limma_results %>%
  mutate(is_poi = ifelse(phosphosite %in% unique(highlight$phosphosite), 'mTORC1 substrate', 'Other')) %>%
  ks.test(abs(logFC) ~ is_poi, data=.)
```

```
##
## Asymptotic two-sample Kolmogorov-Smirnov test
##
## data: abs(logFC) by is_poi
## D = 0.066552, p-value = 0.2464
## alternative hypothesis: two-sided
```

```
phospho_limma_results %>%
  mutate(is_poi = ifelse(UniprotID %in% unique(mtor_substrates_updated$Uniprot.ID), 'mTORC1 substrate',
    'Other')) %>%
  ks.test(abs(logFC) ~ is_poi, data=.)
```

```
##
## Asymptotic two-sample Kolmogorov-Smirnov test
##
## data: abs(logFC) by is_poi
## D = 0.065534, p-value = 0.2044
## alternative hypothesis: two-sided
```

Save results

```
phospho_limma_results %>%
  select(UniprotID, phosphosite, phospho_pos, logFC, P.Value, adj.P.Val,
         Protein.names=Protein.names.y, Gene.Names, Gene.Names.First, site_seq) %>%
  write.table(here('results/PROTAC_phospho_limma_results.tsv'),
             sep='\t', row.names=FALSE, quote=FALSE)
```

Finally, session info:

```
sessionInfo()
```

```
## R version 4.5.2 (2025-10-31)
## Platform: aarch64-apple-darwin20
## Running under: macOS Sonoma 14.7
##
## Matrix products: default
## BLAS: /System/Library/Frameworks/Accelerate.framework/Versions/A/Frameworks/vecLib.framework/Versions/A/
## LAPACK: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRlapack.dylib; LAPACK v
##
## locale:
## [1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8
##
## time zone: Europe/London
## tzcode source: internal
##
## attached base packages:
## [1] stats4 stats graphics grDevices utils datasets methods
## [8] base
##
## other attached packages:
## [1] ggbeeswarm_0.7.3 tidyr_1.3.2
## [3] dplyr_1.2.1 ggrepel_0.9.7
## [5] ggplot2_4.0.3 limma_3.66.0
## [7] here_1.0.2 biomasslmb_0.0.5
## [9] QFeatures_1.20.0 MultiAssayExperiment_1.36.2
## [11] SummarizedExperiment_1.40.0 Biobase_2.70.0
## [13] GenomicRanges_1.62.1 Seqinfo_1.0.0
## [15] IRanges_2.44.0 S4Vectors_0.48.1
## [17] BiocGenerics_0.56.0 generics_0.1.4
## [19] MatrixGenerics_1.22.0 matrixStats_1.5.0
##
## loaded via a namespace (and not attached):
## [1] DBI_1.3.0 rlang_1.2.0 magrittr_2.0.5
## [4] clue_0.3-68 otel_0.2.0 compiler_4.5.2
## [7] RSQlite_3.53.2 systemfonts_1.3.1 png_0.1-9
## [10] vctr_0.7.3 reshape2_1.4.5 stringr_1.6.0
## [13] ProtGenerics_1.42.0 pkgconfig_2.0.3 crayon_1.5.3
## [16] fastmap_1.2.0 backports_1.5.0 XVector_0.50.0
## [19] labeling_0.4.3 rmarkdown_2.31 visdat_0.6.0
## [22] ragg_1.5.0 purrr_1.2.2 bit_4.6.0
## [25] xfun_0.59 cachem_1.1.0 blob_1.3.0
```


## [28] DelayedArray_0.36.1	cluster_2.1.8.2	R6_2.6.1
## [31] stringi_1.8.7	RColorBrewer_1.1-3	genefilter_1.92.0
## [34] Rcpp_1.1.1-1.1	knitr_1.51	BiocBaseUtils_1.12.0
## [37] Matrix_1.7-4	splines_4.5.2	igraph_2.3.2
## [40] tidyselect_1.2.1	rstudioapi_0.19.0	abind_1.4-8
## [43] yaml_2.3.12	lattice_0.22-9	tibble_3.3.1
## [46] plyr_1.8.9	withr_3.0.3	KEGGREST_1.50.0
## [49] S7_0.2.2	evaluate_1.0.5	uniprotREST_1.0.0
## [52] survival_3.8-6	zip_3.0.0	Biostrings_2.78.0
## [55] pillar_1.11.1	corrplot_0.95	checkmate_2.3.4
## [58] rprojroot_2.1.1	scales_1.4.0	xtable_1.8-8
## [61] glue_1.8.1	lazyeval_0.2.3	tools_4.5.2
## [64] robustbase_0.99-7	openxlsx_4.2.8.1	annotate_1.88.0
## [67] XML_3.99-0.23	grid_4.5.2	MsCoreUtils_1.22.1
## [70] AnnotationDbi_1.72.0	beeswarm_0.4.0	vipor_0.4.7
## [73] naniar_1.1.0	cli_3.6.6	textshaping_1.0.4
## [76] S4Arrays_1.10.1	AnnotationFilter_1.34.0	gtable_0.3.6
## [79] DEoptimR_1.2-0	digest_0.6.39	SparseArray_1.10.10
## [82] farver_2.1.2	memoise_2.0.1	htmltools_0.5.9
## [85] lifecycle_1.0.5	httr_1.4.8	statmod_1.5.1
## [88] bit64_4.8.2	MASS_7.3-65	